

Fig. 3: Actual-size PCB pattern of the transmitter section

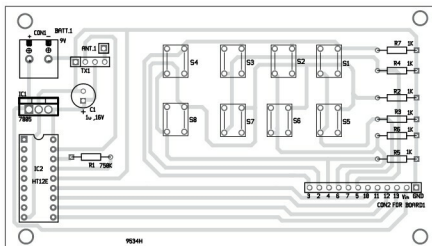


Fig. 4: Component layout of the transmitter section

74HCT154, a 4-to-16 decoder is used to drive the select pins of the voice IC (IC6).

For recording the voice, make REC pin of APR33A3 logic low by closing switch S17. Then, drive message switches (S9-S16) connected to APR33A3 to logic low, sequentially, as per the number of audio messages to be recorded. The chip will play back a beep and the message can then be recorded. When the pin is released, the chip will play back the beep twice to indicate that the message has been recorded.

To play back the message recorded, REC pin should be driven to logic high and then message switches (S9-S16) should be driven from logic high to logic low, and the message play back starts. (LM386 is a low-voltage

audio power amplifier. The voltage gain ranges from 20 to 200.)

Construction and testing

An actual-size, single-side PCB of the transmitter section is shown in Fig. 3 and its component layout in Fig. 4. Similarly, an actual-size, single-side PCB for the receiver section is shown in Fig. 5 and its component layout in Fig. 6.

Assemble the circuit on the PCBs as it minimises time and assembly errors. Carefully assemble the components and double-check for any overlooked error(s). Use proper IC base for the MCU and other ICs.

Message switches at the receiver side and the number of switches at the transmitter side of the keypad may be increased as per requirement,

PARTS LIST

Semiconductors:

- IC1, IC3 - 7805 voltage regulator
- IC2 - HT12E encoder
- IC4 - HT12D decoder
- IC5 - 74HCT154, 4-to-16 line decoder
- IC6 - APR33A3 voice record and play back IC
- IC7 - LM386 low-voltage audio power amplifier
- LED1 - 5mm LED
- TX1 - 433MHz RF transmitter module
- RX1 - 433MHz RF receiver module

- BOARD1 - Arduino UNO board
- BOARD2 - Arduino UNO board

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 750-k Ω -ohm
- R2-R7, R12, R13 - 1-k Ω -ohm
- R8 - 33-k Ω -ohm
- R9 - 47-k Ω -ohm
- R10, R11, R14 - 4.7-k Ω -ohm

Capacitors:

- C1, C2, C9 - 1 μ F, 16V electrolytic
- C3 - 10 μ F, 16V electrolytic
- C4, C5, C10 - 33-k Ω -ohm
- C6, C7, C11 - 100nF ceramic
- C8, C12 - 1nF ceramic

Miscellaneous:

- CON1, CON3 - 2-pin connector
- CON2 - 12-pin connector
- CON4 - 5-pin connector
- CON5 - 6-pin connector
- LS1 - 4-ohms, 0.5W speaker
- MIC1 - Electret condenser microphone
- ANT.1, ANT.2 - 30cm-long single-strand wire antenna
- BATT.1, BATT.2 - 9V battery
- S1-S16, S18 - Tactile switch
- S17 - On/off toggle switch
- USB A-B cable

depending on the number of I/O pins available on Arduino UNO boards.

CON2 is a 12-pin connector used to interface the transmitter side PCB with Arduino UNO (Board1) as per the pins mentioned on the component side as shown in Fig. 4.

Similarly, CON4 is a 5-pin connector and CON5 is a 6 pin connector used to interface the receiver-side PCB with Arduino UNO (Board2) as per the pins mentioned on the component side as shown in Fig. 6.

CON1 and CON3 are 2-pin connectors used to connect 9V battery. To program the MCUs in Arduino boards, use a USB-A-B cable to connect to your system.

For troubleshooting, verify the voltages listed in test points table.

Software

The source code of the transmitter side contains a 2D array for the